into the data by looking at the big picture and analyzing the data from a very high level to draw conclusions about it. We use aggregation APIs to provide analytics in Elasticsearch.

Aggregations fall into three categories:

- *Metric aggregations*—Simple aggregations like `sum`, `min`, `max`, and `avg`. They provide an aggregate value across a set of document data.
- *Bucket aggregations*—Aggregations that collect data into "buckets," segregated by intervals such as days, age groups, and so on. These help us build histograms, pie charts, and other visualizations.
- *Pipeline aggregations*—Aggregations that work on the output from other aggregations.

We go over a few examples of metric and bucket aggregations in the coming pages, but we save the in-depth coverage (including working with pipeline aggregations) until chapter 13.

Similar to searching, we use the `_search` endpoint for aggregations. However, we use a new object called `aggs` (short for `aggregations`) in the request in place of the `query` object we've used so far. To reveal the true power of Elasticsearch, we can combine both search and aggregations in a single query.

To demonstrate aggregations effectively, we need to switch gears and index a new set of data: COVID-related data for 10 countries. The covid-26march2021.txt file to index the data using the _bulk API is available with the book's files; here's a snippet of it.

Listing 2.25 Bulk indexing COVID-related data

```
POST covid/_bulk
{"index":{}}
{"country":"USA","date":"2021-03-26","deaths":561142,"recovered":23275268}
{"index":{}}
{"country":"Brazil","date":"2021-03-26","deaths":307326,"recovered":10824095}
...
```

The `_bulk` API indexes our 10 documents into a newly created `covid` index. As we aren't concerned with the document IDs, we let the system generate a random ID for each of them—hence the empty index name and ID in the `index` action of the API: `{"index":{}})`. Now that we have a set of documents in our covid index, let's carry out some basic aggregation tasks, starting with metric aggregations.

2.6.1 Metrics

Metric aggregations are simple aggregations that we use often in our daily lives: for example, what is the average height of the students in a class? What's the minimum hedge trade? What are the gross earnings of a movie? Elasticsearch provides quite a few such metrics, most of which are self-explanatory.

Before we look at the metrics in action, let's quickly go over the syntax for aggregations. Aggregation queries are written using the same Query DSL syntax as search